Date	31 October 2026
Team ID	XXXXXX
Project Name	Global Fuel Prices and Taxation Analysis (2020–2026)
Maximum Marks	4 Marks

Project Design Phase-I
Solution Architecture

Solution Architecture:
Solution architecture is a complex process — with many sub-processes — that bridges the gap between complex energy business problems and technology solutions. Its goals are to:

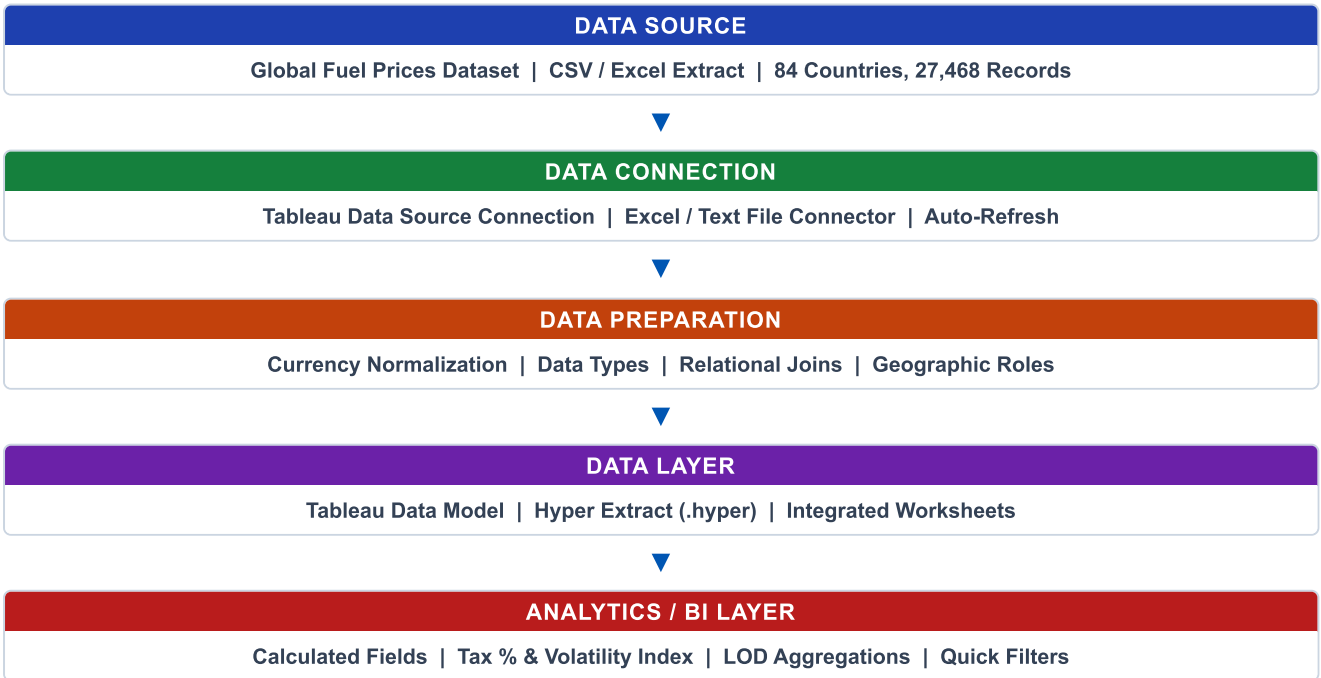
- Find the best tech solution to solve multi-country commodity pricing and taxation challenges.
- Describe the structure, characteristics, behavior, and data transformations of the visual BI suite to project stakeholders.
- Define features, analytical development phases, and technical data pipeline requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Overview:
The architecture connects multi-source global fuel pricing and taxation data through a structured 7-layer Tableau processing framework—moving from raw CSV storage to interactive end-user scenarios for policy, supply chain, and research decision-makers.

SMARTBRIDGE
Let's bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Architecture Diagram



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VISUALIZATION / REPORTING

Scenario 1 (Policy) | Scenario 2 (Logistics) | Scenario 3 (Research) | Interactive Tooltips

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BUSINESS USERS

Policy Consultants | Freight Logistics Directors | Energy Economics Researchers

Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Source	The global fuel dataset holding 27,468 records across 84 countries, tracking Petrol, Diesel, LPG, Tax %, Subsidy Flag, Income Tiers, and Brent crude rates (2020–2026).	Excel / CSV file
Data Connection	Connects Tableau Desktop directly to source spreadsheets, enabling background extract updates whenever new global market pricing data is ingested.	Tableau Desktop Data Source Connection
Data Preparation	Cleans raw records, standardizes currency fields, maps country names to geographic roles, and structures data relations for seamless multi-dimensional slicing.	Tableau Data Source Tab
Data Layer	Stores the optimized packaged data model extract (.hyper) that serves as the single source of truth for all worksheets and story points.	Tableau Data Model / Packaged Extract

Component Name	Description / Role in Architecture	Technology Used
Analytics / BI Layer	Computes key analytical measures (Net Tax Share %, Volatility Index, Brent Crude Correlation Coefficients) using Level of Detail (LOD) expressions and aggregated business logic.	Tableau Calculated Fields & LOD Expressions
Visualization / Reporting	Presents insights via interactive dashboards featuring filled maps, scatter plots, dual-axis trendlines, and cross-tabulations organized across three target stakeholder scenarios.	Tableau Desktop (Scenario Dashboards 1, 2, & 3)
Business Users	Policy Consultants, Supply Chain Directors, and Energy Economists explore dashboards interactively using dynamic quick-filters and parameter controls to drive decision-making.	Tableau Public / Browser-based Dashboard Viewer

Solution Architecture Sign-Off:

Architecture Lead: Project Team	**Verified By:** BI Solutions Architect	**Approved Date:** 31 October 2026